



ONLINE SHOPPING Dataset Analysis

This project analyzes customer behavior in an online shopping platform using Python, Pandas, NumPy, and Matplotlib.



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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('online_shopping.csv')
df.head()
df.describe()
df.info()
plt.figure(figsize=(10,6))
sns.countplot(data=df, x='Purchase')
plt.title('Purchase Distribution')
plt.show()
```





PROJECT OVERVIEW



OBJECTIVE

Analyze customer behavior
in an online shopping platform.

MAIN GOALS



Understand



Clean



Explore



Visualize



Insights



DATASET FEATURES



Age



Gender



Time on Site



Pages Viewed



Device Type



Purchase Amount



Discount Used



Click Rate



Return Customer



TOOLS & TECHNOLOGIES



TECHNOLOGIES USED

PYTHON LIBRARIES:



Pandas
Data handling



NumPy
Numerical operations



Matplotlib
Visualization



DEVELOPMENT ENVIRONMENT:



Jupyter Notebook



PANDAS



NUMPY



MATPLOTLIB



JUPYTER NOTEBOOK



DATA CLEANING



DATA CLEANING PROCESS

STEPS PERFORMED:



Checked missing values



Removed duplicate records



Corrected data types



Identified outliers



Resolved inconsistent values





DATASET OVERVIEW

ROWS

588

ROW

TOTAL

9

COLUMNS

COLUMNS

age

gender

time_on_site

pages_viewed

device

purchase_amount

discount_used

click_rate

return_customer

DATA TYPES

NUMERICAL

Age, TimeOnSite, PagesViewed,
PurchaseAmount, ClickRate, ...

CATEGORICAL

Gender, Device, ReturnCustomer,
DiscountUsed, ...



DATA CLEANING & PREPROCESSING

DATA CLEANING STATISTICS

ORIGINAL ROWS

588

FINAL ROWS

567

MISSING VALUES

239

DUPLICATES REMOVED

21



Clean data is the foundation of **accurate analysis** and **meaningful insights**.

DATA CLEANING DETAILS

A comprehensive overview of the cleaning operations performed on the dataset to ensure high data quality.



CLEANING SUMMARY

Original Rows	588
Final Rows	567
Missing Values	239
Duplicates Removed	21

01

MISSING VALUES

181 numerical → filled with median
58 categorical → filled with mode

02

DATA TYPES

pages_viewed and
return_customer
converted to int64

03

STANDARDIZATION

device, gender, discount_used
labels **standardized**

04

INVALID VALUES

time_on_site and click_rate
negative values **replaced
with median**

05

OUTLIER TREATMENT

click_rate and time_on_site
capped using **IQR method**

06

DUPLICATE REMOVAL

21 duplicate rows
identified and **removed**



FINAL VALIDATION

Final validation confirmed **0 missing values**, **0 duplicates**, and **0 negative values**.



UNIVARIATE ANALYSIS

Understanding Individual Variables



VISUALIZATIONS:



Age Distribution Histogram



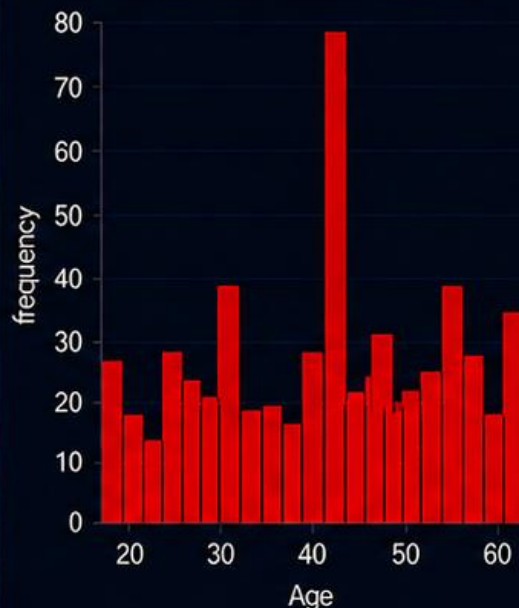
Gender Distribution Histogram



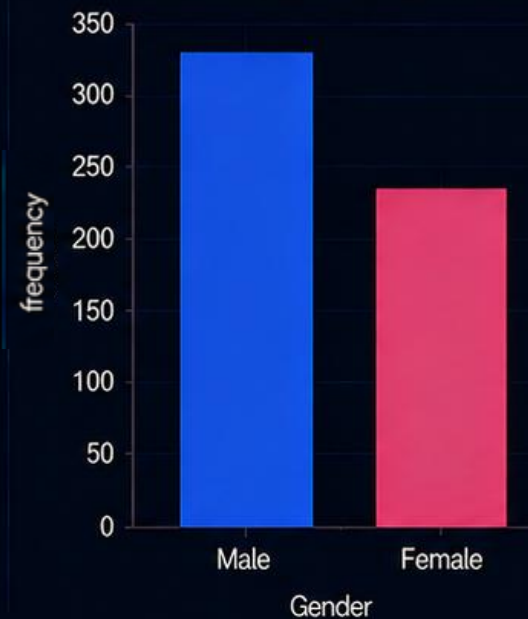
Device Type Histogram



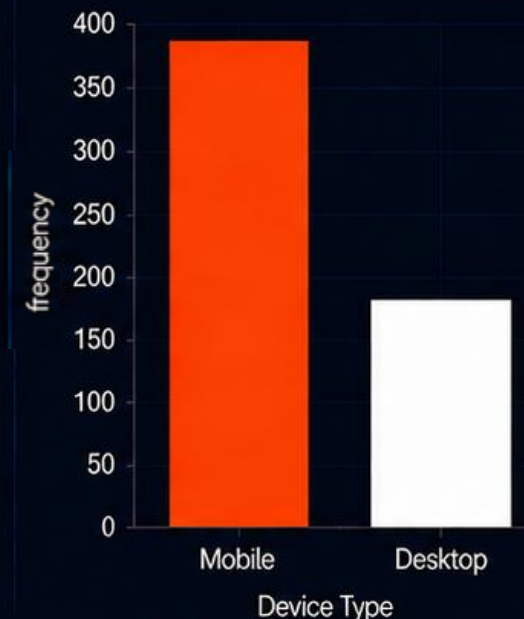
1. AGE DISTRIBUTION Histogram



2. GENDER DISTRIBUTION Histogram



3. DEVICE TYPE Histogram



KEY FINDINGS:



Most users are **between 41–43** years old.



Mobile users **were more** common.



Male use site **common than** FeMale.



UNIVARIATE ANALYSIS

Understanding Individual Variables



VISUALIZATIONS:



Return Customer Distribution Histogram



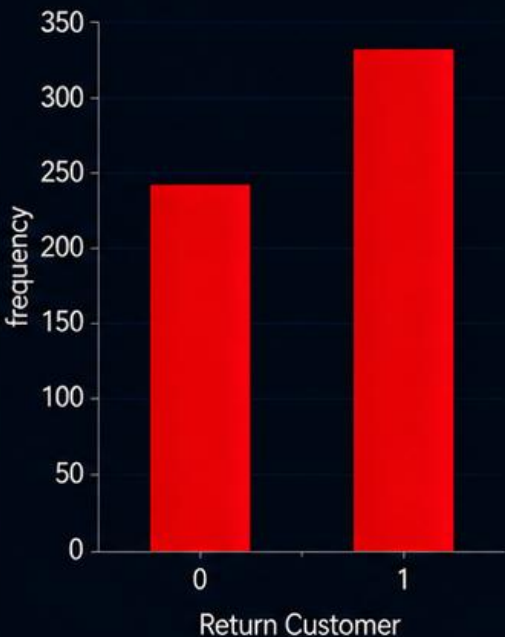
Click Rate Distribution Histogram



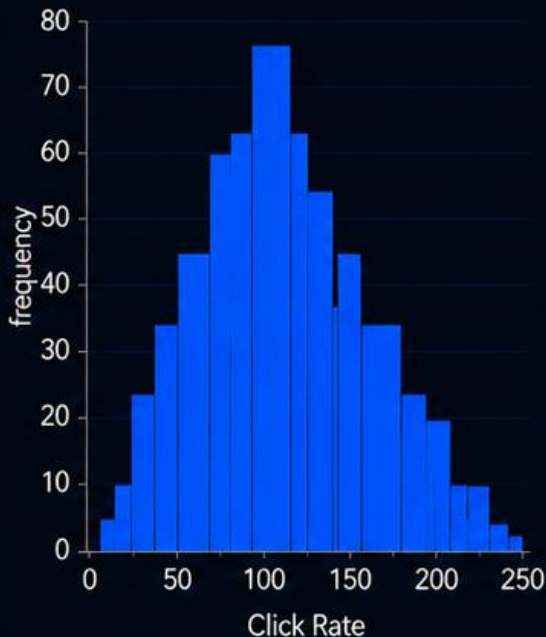
Purchase Amount Distribution Histogram



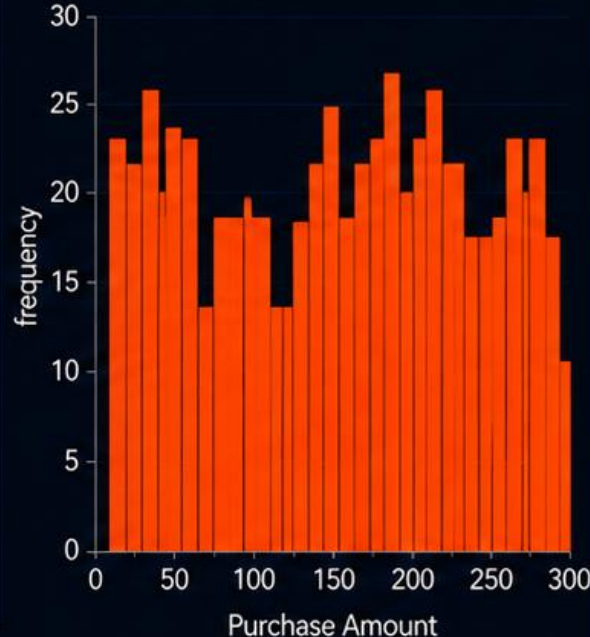
1 RETURN CUSTOMER DISTRIBUTION (HISTOGRAM)



2 CLICK RATE DISTRIBUTION (HISTOGRAM)



3 PURCHASE AMOUNT DISTRIBUTION (HISTOGRAM)



KEY FINDINGS:



A large portion of customers are **return customers**, indicating strong customer retention.



Click rates are concentrated around the **mid range (around 100)**, showing most users have moderate engagement.



Purchase amounts are **evenly spread** across a wide range, indicating no dominant purchase amount.



UNIVARIATE ANALYSIS

Understanding Individual Variables



VISUALIZATIONS:



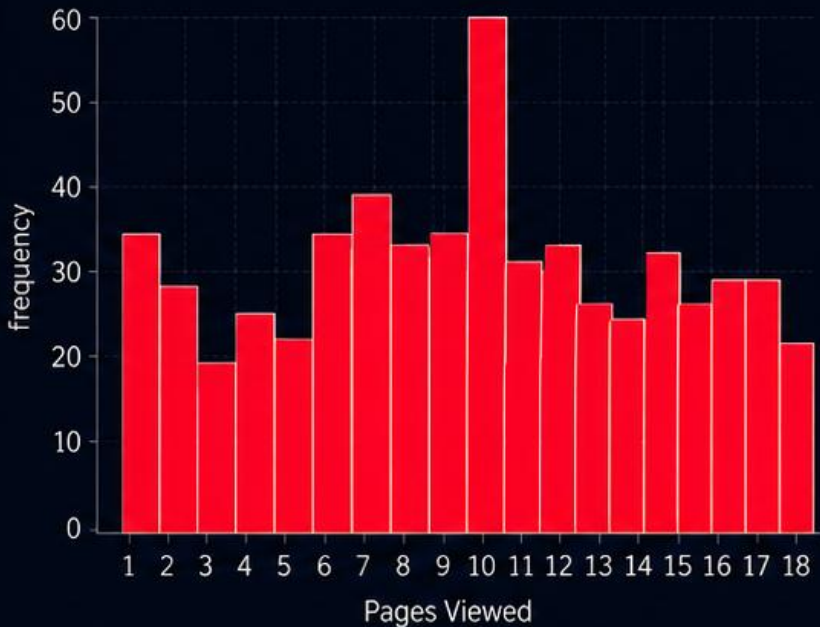
Pages Viewed Distribution Histogram



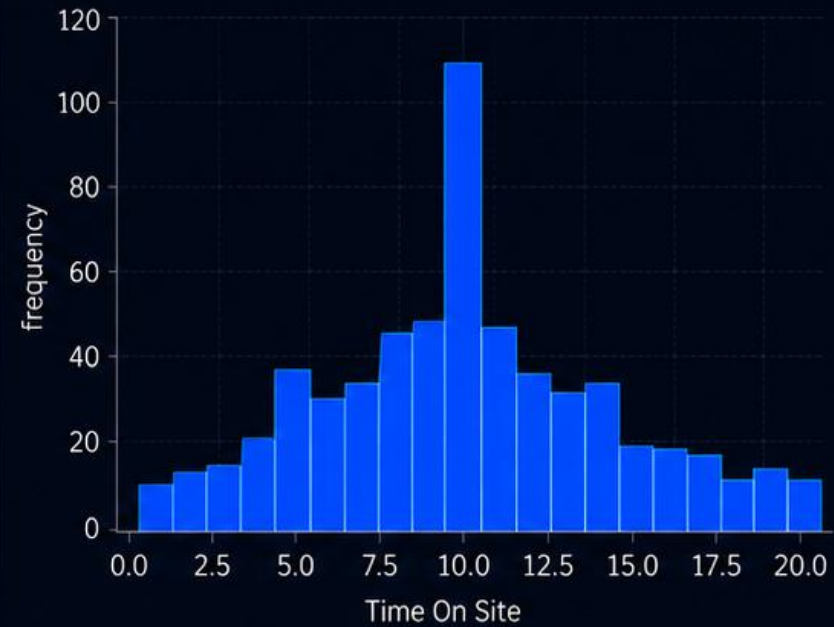
Time On Site Distribution Histogram



1 PAGES VIEWED DISTRIBUTION (HISTOGRAM)



2 TIME ON SITE DISTRIBUTION (HISTOGRAM)



KEY FINDINGS:



Most users view around **8-12 pages**, with the highest frequency at **10 pages**.



Most users spend around **8-12 minutes** on the site, with the peak at **10 minutes**.



UNIVARIATE ANALYSIS

Understanding Individual Variables



VISUALIZATIONS:



Discount Used Distribution
(Bar Chart)

Most customers are split
almost equally between
using and not using a
discount.



1

DISCOUNT USED DISTRIBUTION (BAR CHART)



DATA INSIGHTS

BIVARIATE ANALYSIS

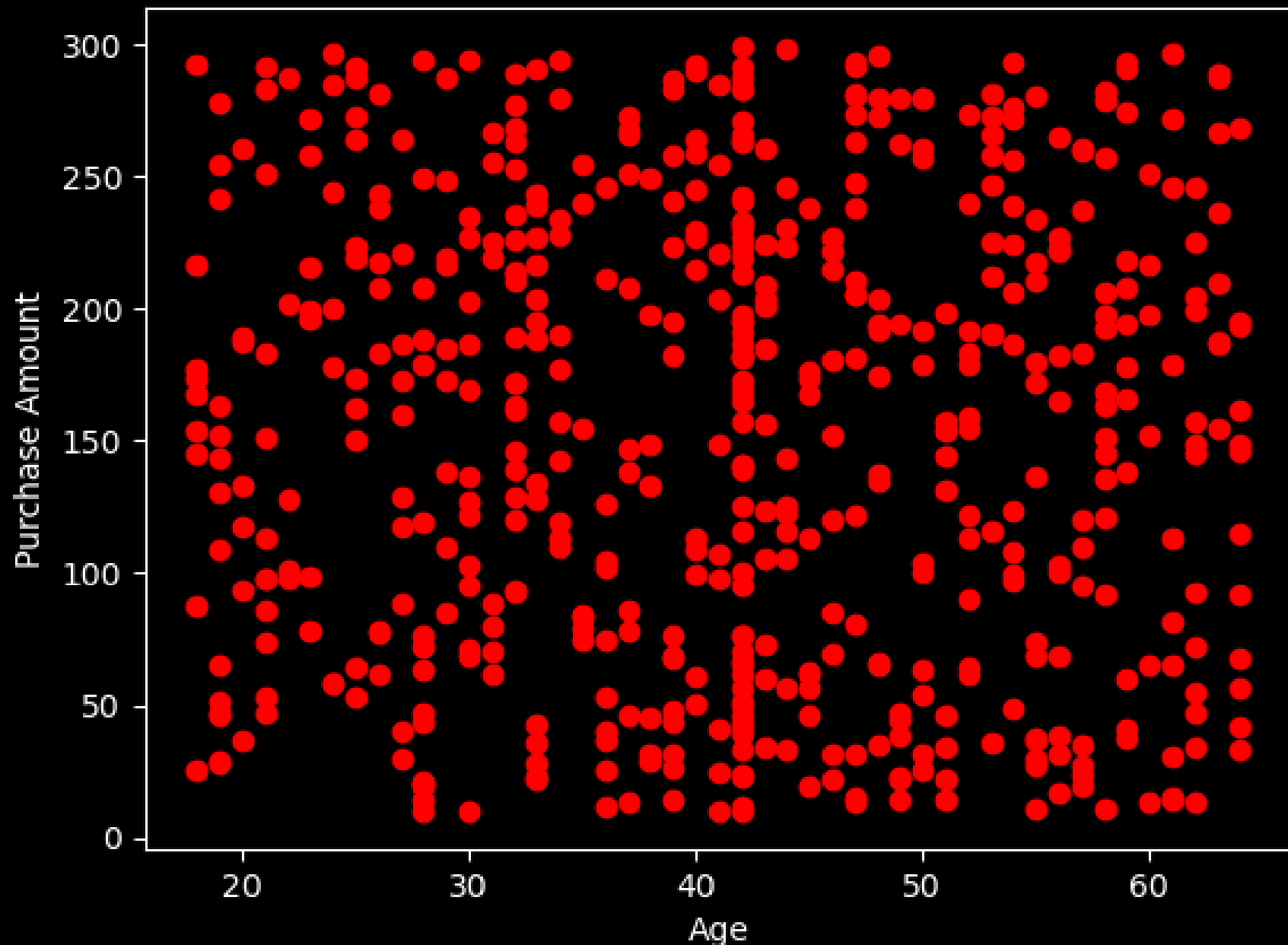
Relationships Between Variables



MAIN INSIGHT

There is no clear
relationship between age
and purchase amount.

Age vs Purchase Amount





DATA INSIGHTS

BIVARIATE ANALYSIS

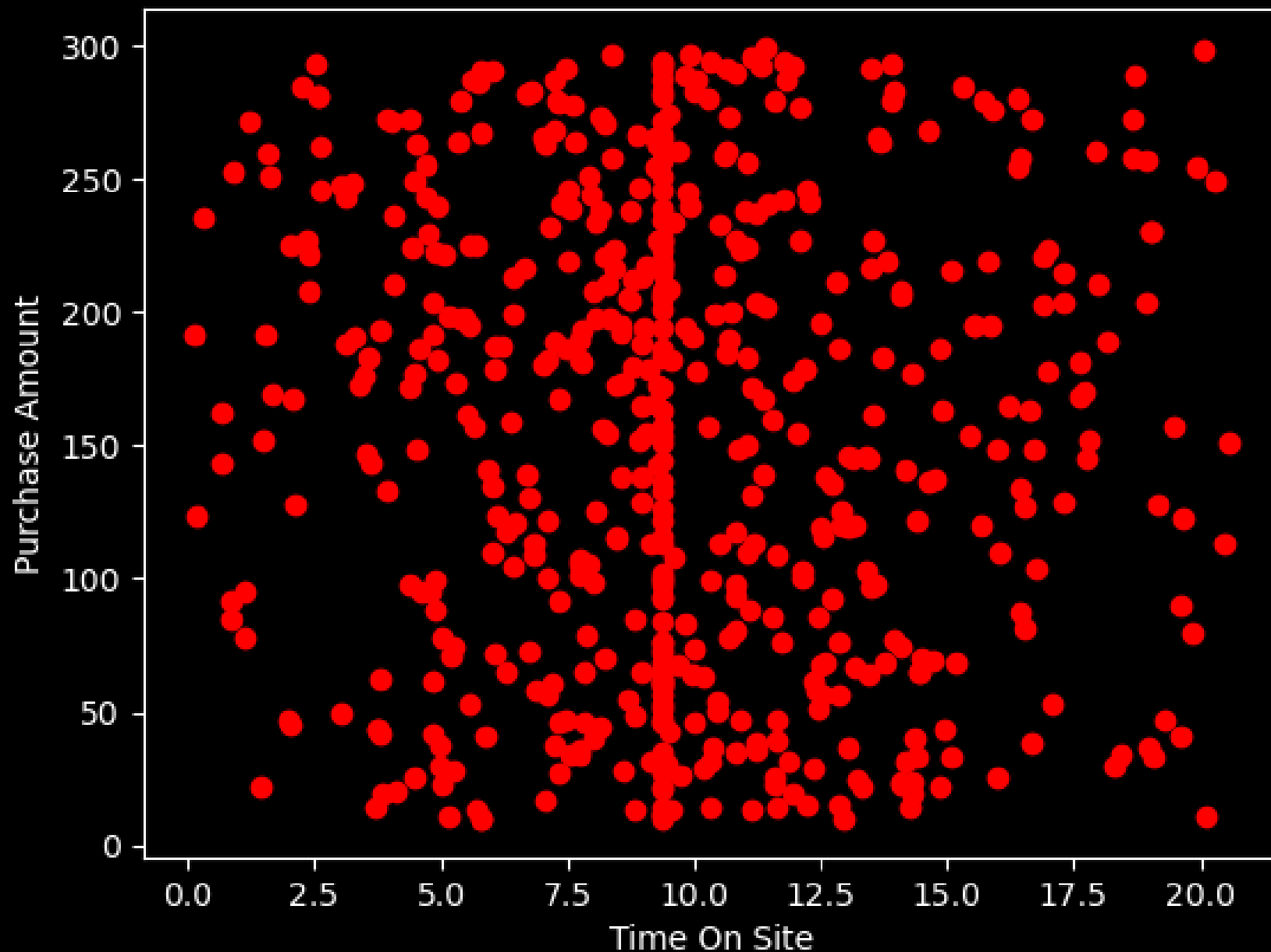
Relationships Between Variables



MAIN INSIGHT

There is no clear
relationship between time
on site and purchase
amount.

Time on Site vs Purchase Amount





DATA INSIGHTS

BIVARIATE ANALYSIS

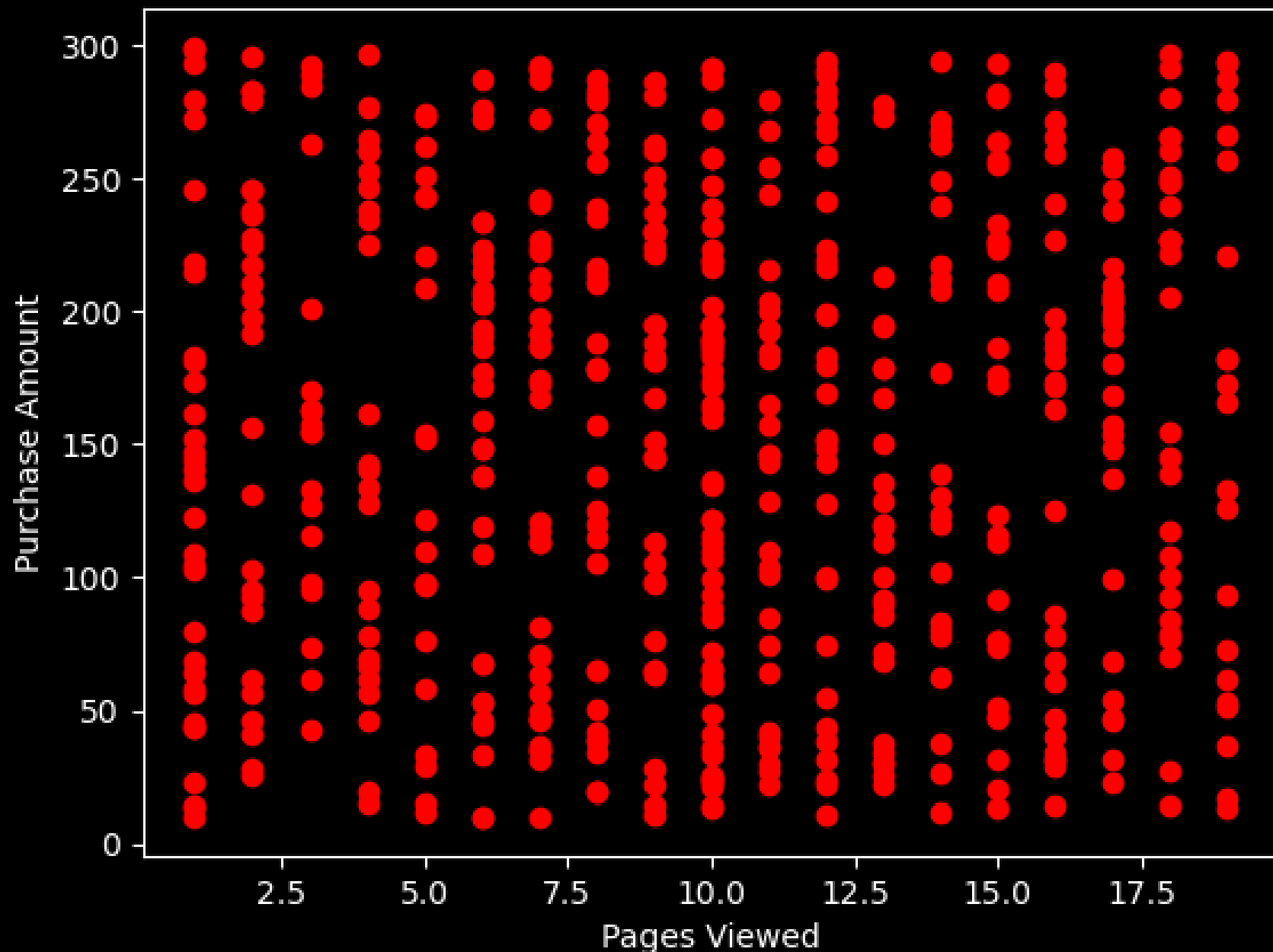
Relationships Between Variables



MAIN INSIGHT

There is no clear
relationship between
pages viewed and
purchase amount.

Pages Viewed vs Purchase Amount





DATA INSIGHTS

BIVARIATE ANALYSIS

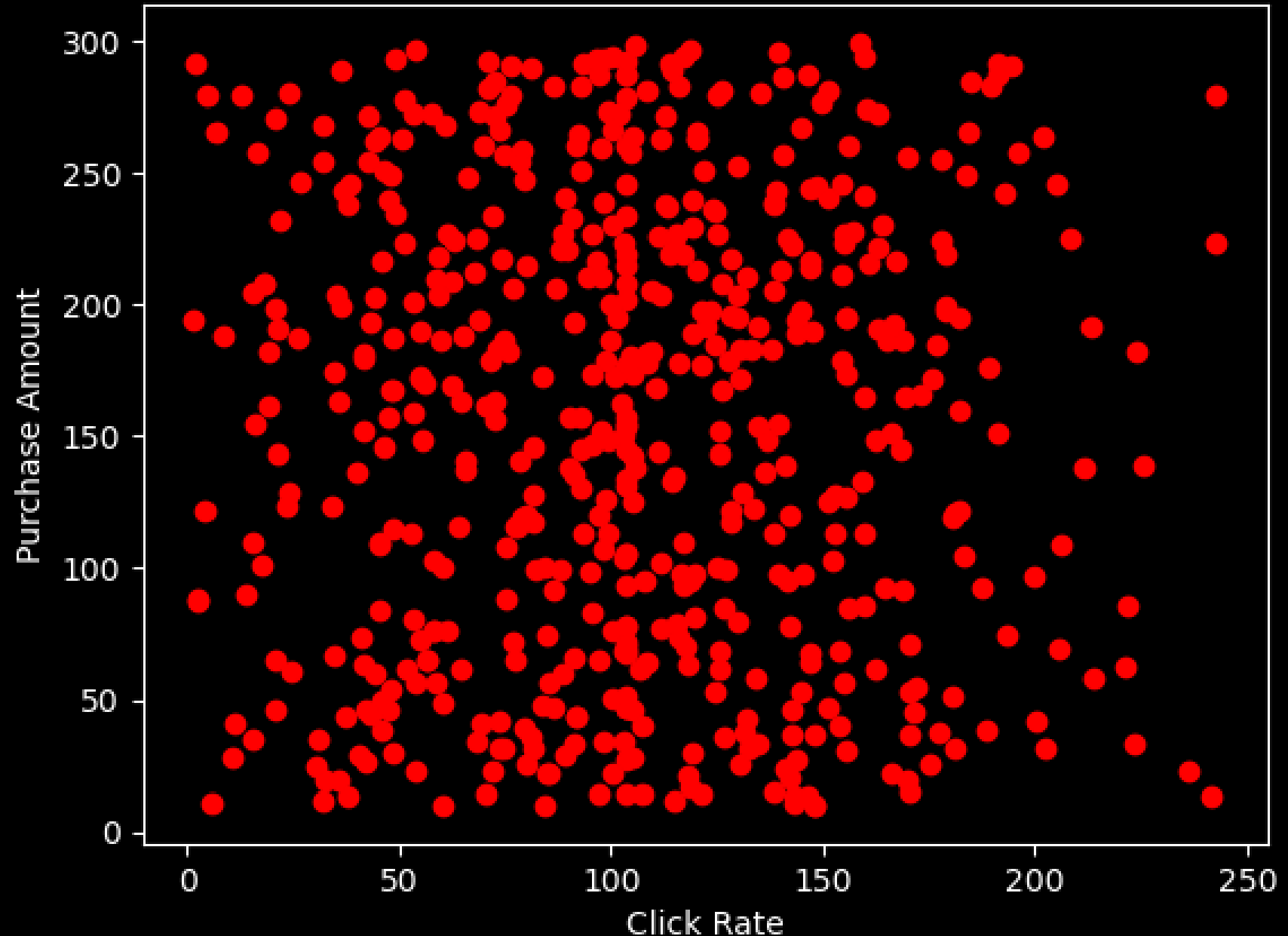
Relationships Between Variables



MAIN INSIGHT

There is no clear
relationship between
click rate and
purchase amount.

Click Rate vs Purchase Amount



BIVARIATE ANALYSIS

Relationships Between Variables



MAIN INSIGHT



Median: $F > M$



Box Height: $F < M$



Total Range:
 $\text{almost } F = M$



BIVARIATE ANALYSIS

Relationships Between Variables



MAIN INSIGHT



Median: $D < M$



Box Height: $D < M$



Total Range:
 $\text{almost } F = M$



BIVARIATE ANALYSIS

Relationships Between Variables



KEY FINDINGS



Median: **NO > YES**



Box Height: **NO > YES**



Total Range:
NO > YES



BIVARIATE ANALYSIS

Relationships Between Variables



MAIN INSIGHT



Median: 0 = 1

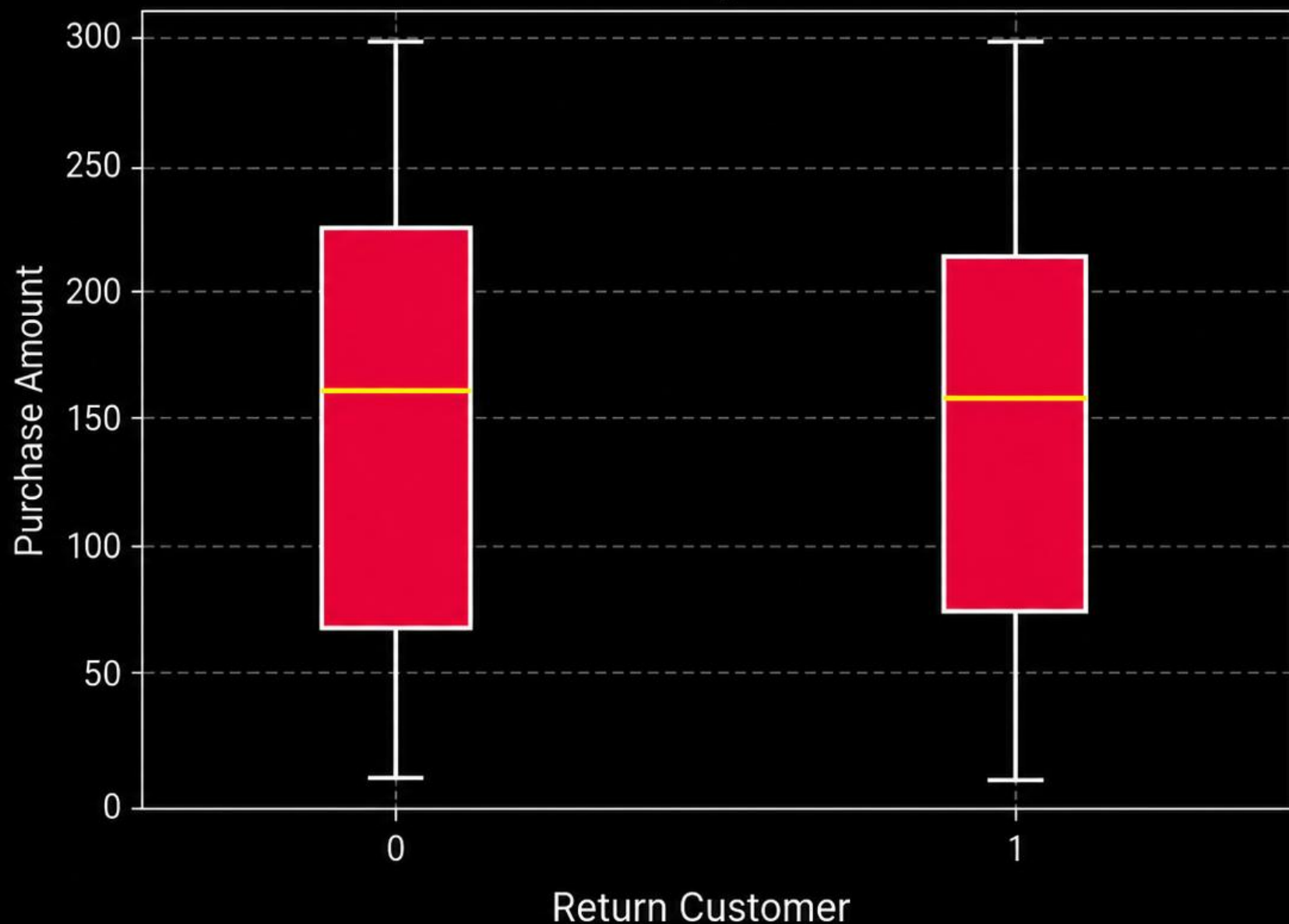


Box Height: 0 > 1



Total Range:
almost 0 = 1

Purchase Amount By Return Customer



BIVARIATE ANALYSIS

Relationships Between Variables



MAIN INSIGHT



The correlation analysis shows mostly **weak relationships** between variables, indicating no strong linear dependencies.

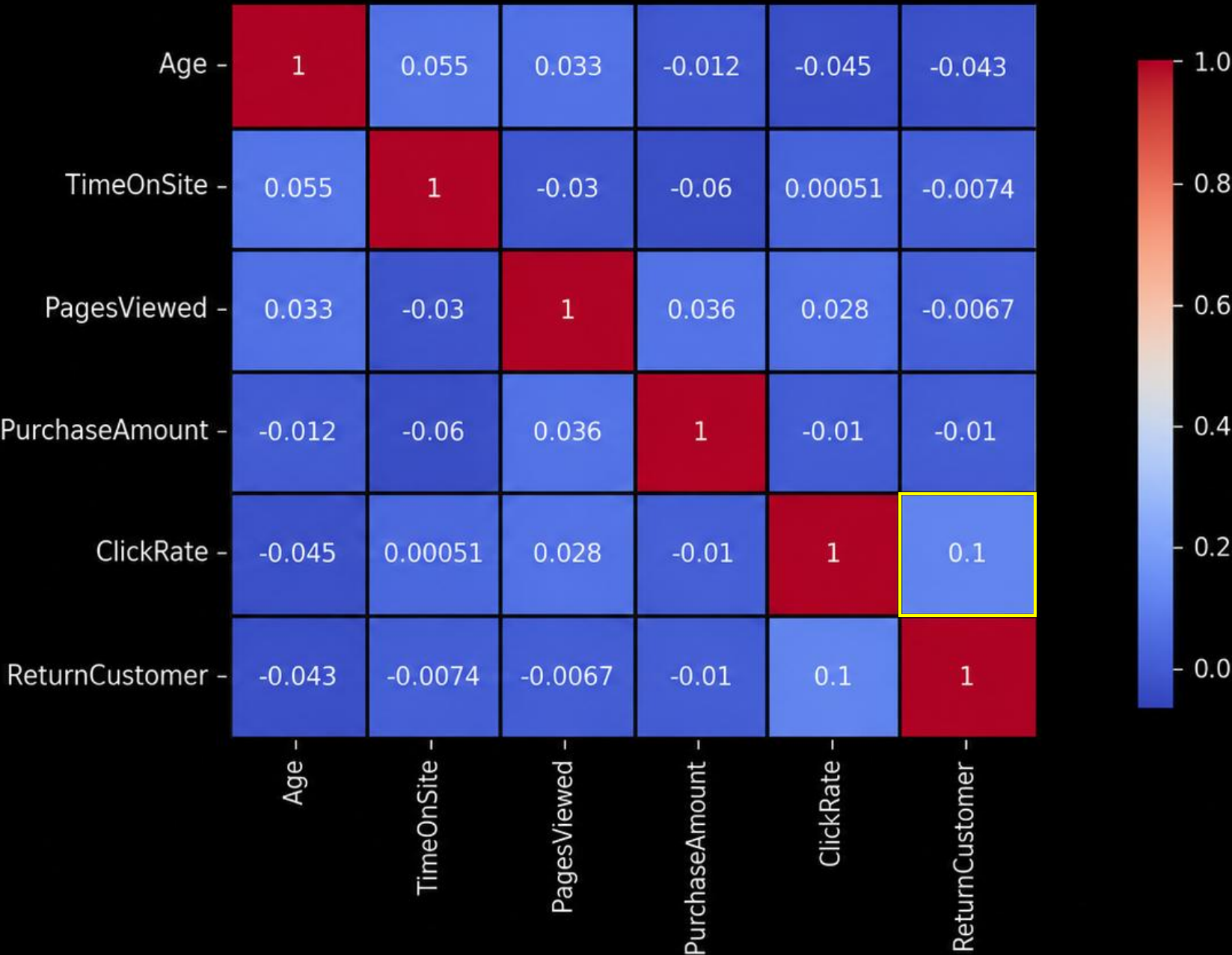


The only slight positive relationship is between **ClickRate** and **ReturnCustomer**,



suggesting that more active users are a bit **more likely to return**.

Correlation Heatmap



BIVARIATE ANALYSIS

Relationships Between Variables



MAIN INSIGHT



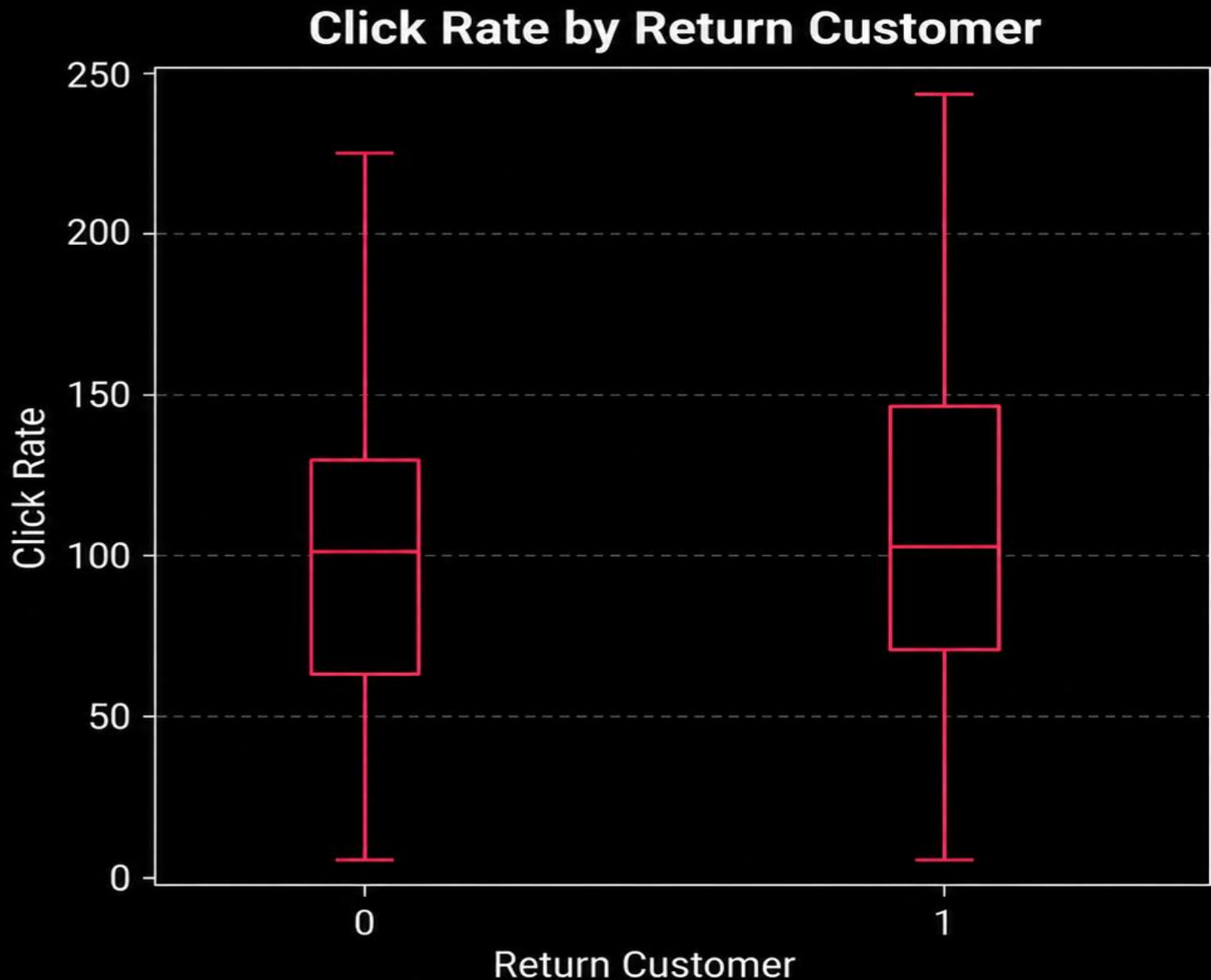
Median: $0 = 1$



Box Height: $0 < 1$



Total Range:
 $0 < 1$



KEY FINDINGS



01

CUSTOMER BEHAVIOR

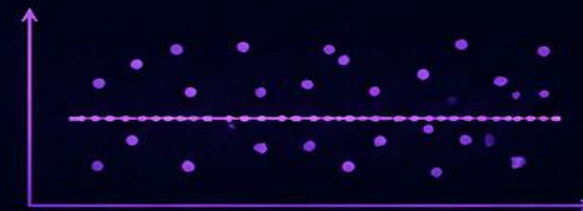
Most users are around **42 years old**, mainly **male**, and prefer **mobile devices**.



02

VARIABLE RELATIONSHIPS

Most variables showed **weak relationships**, meaning no single factor strongly affects purchases.



03

CORRELATION

A slight positive correlation was found between **Click Rate** and **Return Customer**.



04

CONCLUSION

Customer behavior analysis provided **useful insights** into user interaction and purchasing patterns.





THANK YOU!

We appreciate your time and attention.
Questions and feedback are welcome!



THANK YOU



ANY QUESTIONS?



LOOKING FORWARD
TO YOUR FEEDBACK!

